

B.Tech. Degree V Semester (Supplementary) Examination, May 2006

ME 505 HEAT AND MASS TRANSFER (2002 Admissions onwards)

Time: 3 Hours

Maximum Marks: 100

- I a) Explain the following terms:
 (i) Critical insulation thickness
 (ii) Conduction shape factor
 (iii) Lumped capacity
 (iv) Fin effectiveness (2½ x 4 = 10)
- b) It is better to use 10 fins of 5 cm length than 5 fins of 10 cm length. State and prove correctness of the statement. Take properties as follows:
 Diameter of fin = 10mm
 Thermal conductivity = 45 W/m-k
 Heat transfer coefficient = 95 W/m²-k (10)
- OR**
- II a) Prove that the temperature distribution in a body at time t during a Newtonian heating or cooling is given by

$$\frac{T - T_{\infty}}{T_i - T_{\infty}} = e^{-Bi Fo}$$
 (10)
- b) Calculate the maximum current that a 2mm bare aluminium ($K=210 \text{ W/mk}$) wire can carry without exceeding a temperature of 225°C , when exposed in an ambient at 25°C with heat transfer coefficient of $10 \text{ W/m}^2\text{-k}$. Take electrical resistance of aluminium wire as $0.037 \Omega/\text{m}$. (10)
- III a) Explain physical significance of the following dimensionless parameters:
 (i) Reynolds number (ii) Prandtl number
 (iii) Nusselt number (iv) Grashof number (2½ x 4 = 10)
- b) Air at 27°C and 1 atm flows over a flat plate at a speed of 2 m/s . Calculate the boundary layer thickness of a distance of 0.2 m and 0.4 m from the leading edge of the plate. Calculate the mass flow rate, which enters the boundary layer between $x=0.2 \text{ m}$ and $x=0.4 \text{ m}$. The viscosity of air at 27°C is $1.85 \times 10^{-5} \text{ kg/ms}$. Assume unit depth in Z-direction. (10)
- OR**
- IV a) The size of CPU of a personal computer is 40 cm wide, 50 cm deep and 10 cm high. Its top surface is dissipating 25 W to its surroundings at 20°C . Calculate the temperature of the top surface. (10)
- b) One hundred tubes of 12 mm in diameter are arranged in a square array and are exposed to steam at atmospheric pressure. Calculate the mass of steam condensed per unit length of the tubes if the tube wall temperature is maintained at 98°C . Take the properties of water film at mean temperature of $(100+98)/2=99^{\circ}\text{C}$. (10)
- V a) State Wein's law of displacement and prove that $E_{b\lambda}$ will be maximum when $\lambda_m T = 2900 \mu - k$. (10)
- b) A room $5 \text{ m} \times 5 \text{ m} \times 2.5 \text{ m}$ is heated by electric heating provided in the ceiling. If the ceiling surface temperature is maintained at 45°C and the temperature of the wall is 25°C in equilibrium condition, find the total heat loss from the ceiling by radiation.
 Emissivity of ceiling = 0.75
 Emissivity of wall = 0.65
 Assume the floor is non-sensitive to radiation. (10)
- OR**

(Turn Over)

- VI a) Sun emits maximum radiation at $\lambda = 0.52\mu$. Assuming Sun as black body, find the surface temperature of the Sun and emissive power at that temperature. (6)
- b) Two large parallel planes with emissivities 0.4 ($T=500\text{K}$) and 0.8 ($T=700\text{K}$) exchange heat. Find the net heat radiated by them and percentage reduction in heat transfer when polished aluminium radiation shield ($\varepsilon = 0.04$) is placed between them. (14)
- VII a) Derive for parallel flow heat exchanger,
- $$\varepsilon = \frac{1 - \exp[-NTV(1+C)]}{1+C} \quad \text{where}$$
- $$C = \frac{C_{\min}}{C_{\max}} = \frac{(\dot{m}C_p)_{\min}}{(\dot{m}C_p)_{\max}} \quad (10)$$
- b) A chemical having specific heat of 3.3kJ/kg.k at a rate of $20,000\text{kg/hr}$ enters a parallel flow heat exchanger at 120°C . The flow rate of cooling water is $50,000\text{ kg/hr}$ with an inlet temperature of 20°C . The heat transfer area is 10m^2 and overall heat transfer coefficient is $1050\text{W/m}^2\text{K}$. Find:
- the effectiveness of the heat exchanger
 - outlet temperature of water and chemical.
- Take C_p of water as 4.186kJ/kg.k (10)
- OR**
- VIII a) Explain shell and tube type heat exchanger. Why baffles are used? (8)
- b) Assume the condenser of a large power plant to be a shell and tube type heat exchanger consisting of a single shell and 30,000 tubes, each executing two passes. The tubes are of thin wall constructed with 25mm in diameter, and steam condenses on their outer surface with an associated convection coefficient of $11,000\text{W/m}^2\text{K}$. The heat transfer rate that must be effected by exchanger is 2000 MW and this is accomplished by cooling water through the tube at the rate of $30,000\text{ kg/s}$. The water enters at 20°C , while steam condenses at 50°C . What is the temperature of cooling water coming out? What is the required tube length per pass? (12)
- IX a) Define and explain the physical significance of:
- Schmidt number
 - Lewis number
 - Sherwood number
- (9)
- b) The tyre tube of a vehicle has a surface area of 0.62m^2 and wall thickness of 10mm . The tube has air filled in it at a pressure of 2.2 bar . The air pressure drops to 2.18 bar in period of 6 days. The solubility of air in the rubber is 0.075m^3 of air/ m^3 of rubber at 1 bar . The volume of air in the tube is 0.034m^3 and operating temperature is 300K . Calculate the diffusivity of air in the rubber. (11)
- OR**
- X a) An open pan 20 cm in diameter, 20mm deep is filled with water to a level of 10mm and is exposed to air at 25°C . Assuming the mass diffusivity of $0.25 \times 10^{-4}\text{m}^2/\text{s}$, calculate the time require for all the water to evaporate. (10)
- b) Explain equimolar counter diffusion. (10)

